Hyperspectral Anomaly Detection via Image Super-Resolution Processing and Spatial Correlation

Zhuang Li, *Student Member, IEEE*, and Ye Zhang, *Member, IEEE*

***Abstract*— Anomaly detection is a key problem in hyperspectral image (HSI) analysis with important remote sensing applications. Traditional methods for hyperspectral anomaly detection are mostly based on the distinctive statistical features of the HSIs. However, the anomaly-detection performance of these methods has been negatively impacted by two major limitations: 1) failure to consider the spatial pixel correlation and the ground-object correlation and 2) the existence of the mixing pixels caused by both lower spatial resolution and higher spectral resolution, which leads to higher false-alarm rates. In this article, these two problems are largely solved through a novel hyperspectral anomaly-detection method based on image super-resolution (SR) and spatial correlation. The proposed method encompasses two innovative ideas. First, based on the spectral variability in the anomaly targets, an extended linear mixing model can be obtained with more accurate ground-object information. Then, image SR is used to improve the spatial resolution of the HSIs by injecting the ground-object information from the mixing model. This alleviates the effect of mixed pixels on anomaly detection. Second, spatial correlation is exploited jointly with the global Reed-Xiaoli (GRX) method and the ground-object correlation detection for anomaly detection. Experimental results show that the proposed method not only effectively improves the hyperspectral spatial resolution and reduces the false-alarm rate but also increases the detectability with the spatial correlation information. Furthermore, the results for the real HSIs demonstrate that the proposed method achieves higher rates of anomaly detection with lower false-alarm rates.**

***Index Terms*— Anomaly detection, hyperspectral image (HSI), spatial correlation, spectral variability, super-resolution (SR).**

I. Introduction

HYPERSPECTRAL images (HSIs) are significantly important for anomaly detection, as they represent a rich source of continuous spectral information about the materials in a scene [1]–[4]. With the continuous development of the imaging spectrometers, their spectral resolution and real-time performance characteristics have been growingly improved [5].

Nowadays, a HSI has hundreds of bands and the spectral resolution reaches the nanometer level. In addition, such an image can combine 1-D spectral information with 2-D spatial information, leading to high feature details. Therefore, hyperspectral remote sensing is an important means of ground-object detection [6], [7].

One key task in hyperspectral remote sensing is hyperspectral target detection, which can be supervised or unsupervised. In unsupervised target detection or anomaly detection, the target is detected without prior knowledge of the target information [8]. The main purpose of anomaly detection is to detect unusual targets in the HSIs, such as military, man-made, hidden, or camouflaged targets [9]. In essence, hyperspectral anomaly detection is a binary classification problem, in which every image pixel is labeled as an anomaly target or a background pixel [10]. Anomaly detection makes realistic assumptions, as it requires no prior target information, and is hence an important part in hyperspectral target detection. Anomaly detection has been employed in numerous applications such as mineral detection [11], fine classification, environmental monitoring, and food and drug quality detection. However, due to the lack of prior spectral information of the anomaly targets, achieving good anomaly-detection performance with traditional methods is quite difficult [12].

For the last few decades, many methods have emerged for hyperspectral anomaly detection. The most classical algorithm is the Reed–Xiaoli (RX) algorithm proposed by Reed and Yu [13]. This algorithm is essentially based on the generalized likelihood ratio test, which depends on the difference between the background and the target pixels in terms of the mean and variance statistics. However, the RX algorithm makes the unrealistic assumption that the background follows a multivariate Gaussian distribution. As a result, the RX algorithm has a low detection rate, as well as too many false alarms and missed detections. Hence, this algorithm cannot meet the needs of practical applications [14]. Therefore, many improved variants of the RX algorithm have emerged. Kwon and Nasrabadi [15] proposed a nonlinear kernel RX (KRX) method, which maps the images to the high-dimensional feature spaces by the nonlinear kernel functions. This method can fully describe the complex background distribution and effectively reduce the false-alarm rate. Another RX variant, namely, the subspace RX (SSRX) algorithm, maps the spectral information into a linear subspace and, thus, exhibits an increased detection rate [16]. In addition, more recent RX variants (such as the weighted RX (W-RX) [17], iterative

Manuscript received December 23, 2019; revised April 27, 2020; accepted May 25, 2020. Date of publication July 14, 2020; date of current version February 25, 2021. This work was supported in part by the National Science Foundation of China under Grant 61871150. *(Corresponding author: Ye Zhang.)*

The authors are with the Department of Information Engineering, School of Electronics and Information Engineering, Harbin Institute of Technology, Harbin 150001, China (e-mail: lz960505_hit@163.com; zhye@hit.edu.cn).

Color versions of one or more of the figures in this article are available online at https://ieeexplore.ieee.org.

Digital Object Identifier 10.1109/TGRS.2020.3005924

0196-2892 © 2020 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 04:01:16 UTC from IEEE Xplore. Restrictions apply.

linear RX [18], and segmented RX [19] algorithms) produce good simulations of the real image background distributions. In addition, Li *et al.* [20] proposed a transferred deep convolutional neural network for anomaly detection (CNND) in the hyperspectral imagery. In the CNND framework, a reference data set was employed to train the CNN with pixel pairs generated from the labeled samples.

In order to avoid dealing with data distributions, many methods followed different anomaly-detection approaches, such as those based on sparse representations [21]. In recent years, the sparse-signal-representation theory has been rapidly developed. In particular, sparse representations of the HSIs have been used in several applications. Most notably, a collaborative-representation-based detector (CRD) [22] exploits the fact that each background pixel can be approximately represented by its spatial neighborhoods, while anomalies cannot have such representations. The collaboration of the representations is reinforced by the l_2-norm minimization of the representation weight vector.

The low-rank sparse-reconstruction theory decomposes a HSI into sparse and low-rank parts [23]. The background usually shows low-rank characteristics, while the anomaly has sparse characteristics [24]. This characterization can be exploited to achieve anomaly detection. For example, Ning *et al.* [25] achieved hyperspectral anomaly detection by decomposing a HSI into low-rank (or background), sparse (or anomaly), and noise components. Under the constraints of the joint-collaborative-representation (JCR) model, the main component of a HSI is divided into several image blocks that are classified to establish a background dictionary and an anomaly dictionary. The anomaly-detection result is the coefficient associated with the anomaly dictionary. Qu *et al.* [26] proposed a hyperspectral anomaly-detection method based on spectral unmixing and dictionary construction with low-rank decomposition. In this method, the background coefficients are obtained via the low-rank constraints, and the noise component in the anomaly can be removed to reduce effectively the false-alarm rate.

Apart from the earlier methods, this article proposes a hyperspectral anomaly-detection method based on super-resolution (SR) and spatial correlation. The proposed enhancements are inspired by the physical characteristics of the HSIs. One of these key characteristics is that the spatial resolution is low, while the spectral resolution is high. This leads to the mixed-pixel phenomenon [27], in which the spectral content of a single pixel contains different ground objects. This phenomenon negatively impacts the performance of the anomaly-detection methods. Higher spectral resolution alleviates this problem and allows spectral unmixing [28]. The SR methods can be effectively applied for the computational enhancement of the spatial image resolution when the hardware conditions cannot meet the application requirements [29]. In HSI SR, subpixel positioning and mapping can be applied to the input image, and the spectral ground-object information obtained by unmixing can be fully used [30]. The hyperspectral SR processing can improve the spatial image resolution to a certain extent, reduce the impact of the mixed pixels on anomaly detection, and also inject the redundant spectral information of the input spectral image into the HSI with high spatial resolution. These benefits enable full mining of the abnormal target information, boost the anomaly-detection performance, and represent the key contributions of this article. Moreover, spatial correlation [31] in the HSIs can be categorized into two types. The first type is the correlation between the pixels, i.e., the correlation between the tested pixels and the surrounding pixels. The second type is the ground-object correlation, i.e., the correlation between the tested pixels and the surrounding background objects. The anomaly-detection results can be improved by accounting for the two correlation types. In summary, the main contributions of this work are as follows.

1) The extended linear mixing model (ELMM) is used to account for spectral variability and fully reflect the spectral anomaly variations [32]. In addition, HSI SR is introduced for anomaly detection. The spectral information is injected through SR along with spatial correlation. This leads to the doubling of the spatial resolution of the HSIs. This article generally promotes unmixing and SR for hyperspectral anomaly detection. This is conducive to the development and application of the intelligent techniques for HSI analysis.
2) For anomaly detection in the HSIs of high spatial resolution, spatial correlation is used in conjunction with the global RX (GRX) [33] algorithm and the ground-object correlation. This approach can fully protect and enhance the small targets, effectively improve the detection rate, and reduce the false alarms.

The remainder of this article is organized as follows. The details of our method are given in Section II. The experimental results and discussions are reported in Section III. Finally, the conclusions are made in Section IV.

II. Proposed Method

As shown in Fig. 1, the proposed approach consists of four steps. First, on the basis of spectral variability in the HSIs, high-precision abundance maps are obtained by extending the linear mixing model. All kinds of such maps can fully reflect the spectral changes of the anomalous objects. Second, based on the constructed abundance maps, a cellular automata (CA) method is used in subpixel localization and mapping for all kinds of ground objects. Using the subpixel information and the spatial correlation, preliminary superresolved images are obtained. Third, the spatial resolution of each original low-resolution image is improved by the algorithm of hyperspectral projection onto convex sets (POCS). Thus, the HSIs of higher spatial resolution are obtained. Fourth, anomaly detection is performed on the superresolved HSIs using collaborative spatial correlation.

A. Spectral Variability and the Extended Linear Mixing Model

In the HSIs, the spectrum measured at a single pixel is usually a mixture of several features' types, and the linear mixing model can describe the mixing process [34]. The linear

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 04:01:16 UTC from IEEE Xplore. Restrictions apply.

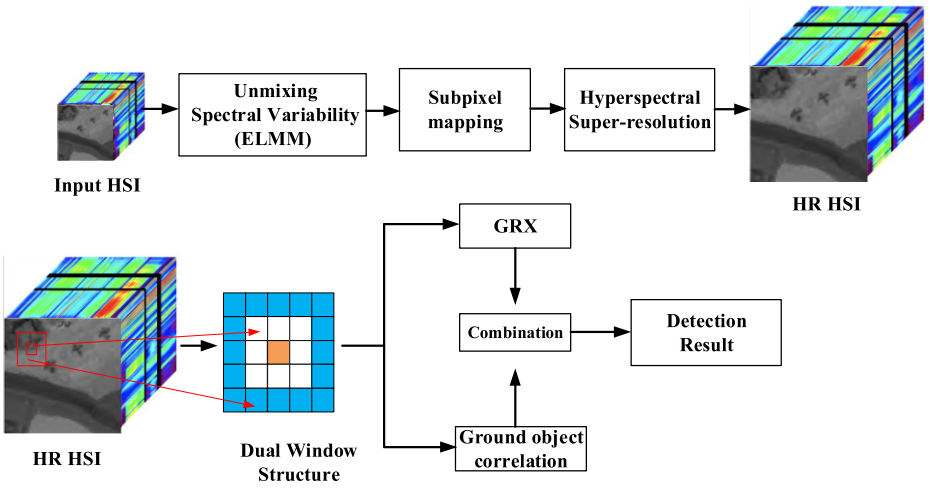


Fig. 1. Overview of the proposed SRP-SC method for anomaly detection.

mixing model is typically expressed as

$$\mathbf{X} = \mathbf{SA} + \mathbf{E} \tag{1}$$

where $\mathbf{X}$ is a HSI of L bands and N pixels, $\mathbf{S}$ is the corresponding endmember matrix, $\mathbf{A}$ is the endmember abundance in the original image, and $\mathbf{E}$ is a noise or the error term.

For each pixel, the linear mixing model in (1) translates to

$$\mathbf{x}_k = \sum_{p=1}^{p} \mathbf{a}_{pk}\mathbf{s}_p + \mathbf{e}_k \tag{2}$$

where $\mathbf{x}_k$ represents the kth pixel in the HSI, $\mathbf{s}_p$ is the endmember spectrum of $\mathbf{x}_k$, P is the number of endmembers in $\mathbf{x}_k$, $\mathbf{a}_{pk}$ is the corresponding abundance vector, and $\mathbf{e}_k$ is the kth noise component. According to the physical characteristics of the endmembers, the abundance vectors must satisfy the nonnegativity and the unit-sum conditions, that is

$$\mathbf{a}_{pk} \geq 0 \tag{3}$$

$$\sum_{p=1}^{P} \mathbf{a}_{pk} = 1. \tag{4}$$

However, the linear mixing model ignores the spectral variability and cannot reflect well the target information details, which are particularly important in anomaly detection. Therefore, for better anomaly-detection performance, the existing linear mixing model should be extended to account for these information details.

Therefore, we use in this article the unmixing method proposed by Drumetz *et al.* [34] to reflect the anomaly target details. Under the linear model assumptions, the ELMM accounts for the endmember spectral variability through a scale factor ψ as follows:

$$\mathbf{x}_k = \sum_{p=1}^{p} \mathbf{a}_{pk} f_{pk}(\mathbf{s}_p) + \mathbf{e}_k \tag{5}$$

where $f_{pk} = \boldsymbol{\psi}_{pk}\mathbf{s}_{0_p}$, $\boldsymbol{\psi}_{pk}$ represents a scale factor, and $\mathbf{s}_{0_p}$ indicates the supplied or extracted reference endmembers

$$\mathbf{x}_k = \sum_{p=1}^{p} \mathbf{a}_{pk}\boldsymbol{\psi}_{pk}\mathbf{s}_{0p} + \mathbf{e}_k = \mathbf{S}_0\boldsymbol{\psi}_{pk}\mathbf{a}_k = \mathbf{S}_0(\boldsymbol{\psi}_k \circ \mathbf{a}_k) + \mathbf{e}_k \tag{6}$$

where the symbol $\circ$ represents the Schur–Hadamard product.

Equation (6) might be rewritten succinctly in a matrix form as

$$\mathbf{X} = \mathbf{S}_0(\boldsymbol{\Psi} \circ \mathbf{A}) + \mathbf{E}. \tag{7}$$

Typically, each of (6) and (7) can be formulated as a nonconvex optimization problem, which can be solved by employing a variable splitting strategy. Specifically, the optimization problem formulation is given by [34]

$$\{\mathbf{X}, \mathbf{S}, \mathbf{A}_k\} = \arg\min_{\mathbf{X},\mathbf{S},\mathbf{A}_k} \sum_{k=1}^{N} \left(\|\mathbf{x}_k - \mathbf{S}_k\mathbf{a}_k\|_2^2 + \lambda_S \|\mathbf{S}_k - \mathbf{S}_0\boldsymbol{\psi}_k\|_F^2 \right) \tag{8}$$

where $\mathbf{S}_0$ is the reference endmember matrix, $\{\mathbf{S}_k\}$ is the set of pixel-dependent endmember matrices, and λ_S denotes a penalty parameter to balance the influence of the two terms. Therefore, we can iteratively optimize the individual variables by the alternating the nonnegative least-square (ANLS) method [35].

The ELMM method can obtain endmember spectral and abundance maps with more accurate and detailed information of the anomaly targets. In this work, the anomaly-detection outcomes of different unmixing methods are compared to verify that the ELMM method can adequately and more accurately extract the anomaly target details. Further discussions on this point are given in Section II-D.

B. Subpixel Localization and Mapping

The composition and abundance of each ground object in the mixed pixels can be obtained through the unmixing methods. However, the spatial distribution of the ground objects in the mixed pixels is generally unknown. Thus, a key problem here is subpixel localization and mapping [36].

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 04:01:16 UTC from IEEE Xplore. Restrictions apply.

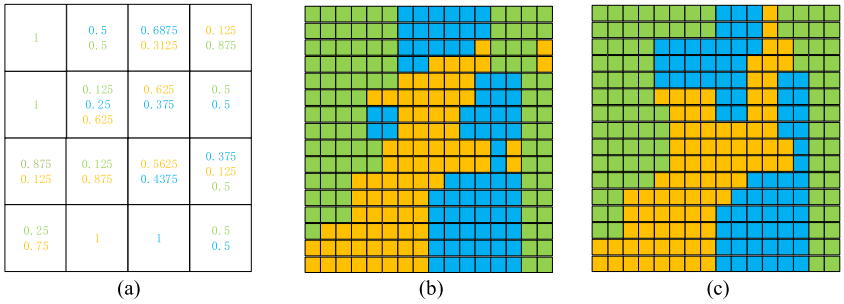


Fig. 2. Relevance of the ground objects for a simple example. (a) Abundance maps with three ground objects are extracted for a 4 × 4-pixel image. (b) One possible subpixel distribution in an image of higher spatial resolution such that each coarse pixel is divided into 4 × 4 subpixels. (c) Another perceived subpixel distribution.

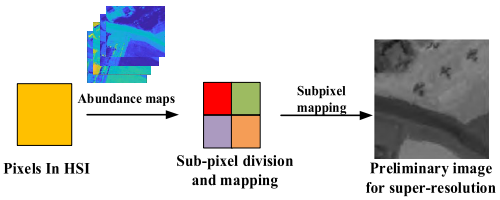


Fig. 3. Subpixel localization and mapping.

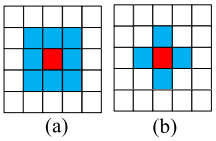


Fig. 4. Types of neighborhood models in cellular automata. (a) Moore type. (b) von Neumann type.

Subpixels result from the cutting of the pixel cells into smaller units and the distribution of the ground object cover information among these subpixels. Specifically, after the unmixing of the original HSIs to obtain the endmember components and abundance maps of the mixed pixels, the original mixed pixels are divided into smaller units or subpixels, for which the proportions of the endmember components of the original pixels are known from unmixing. For subpixel mapping, the subpixels are assigned to the ground-object components. Finally, the spatial distribution of each endmember component in the mixed pixels is obtained [37]. These steps lead to better image details and improved classification accuracy in remote sensing, as shown in Fig. 3.

Subpixel localization is initialized through spatial correlation. Let the original HSI have N different types of ground objects, and let the proportions of various ground objects in each mixed pixel be obtained by unmixing [38]. On this basis, each pixel with low spatial resolution in the original HSI is divided into 2 × 2 subpixels, and the number of subpixels occupied by each type of ground objects can be determined, as shown in Fig. 2. The subpixel localization maximizes the spatial correlation of the ground-object distribution under the premise of satisfying the proportions of the various types of ground objects in the pixel of interest. As shown in Fig. 2, the mixing information of the three types of ground materials is obtained through unmixing, based on spatial correlation. The typical ground-object-distribution maps are as shown in Fig. 2(b) and (c) [39]. Compared with the maps of the ground objects, the map in Fig. 2(c) is more reasonable.

Many methods have been proposed for subpixel localization. Based on CA modeling [40], we proposed a variant of the conventional cellular automaton model for subpixel localization in the HSIs. This variant is called the hyperspectral cellular automaton model.

CA is a discrete system composed of a large number of individuals that are connected through local connections in time, space, and state. The first such system was proposed by von Neuman and is mainly used for the self-replication function of the life-simulation systems in artificial life. After years of development, CA have gradually evolved into a discrete tool for the applications of complex system simulation, such as flood evolution, land use evolution, rainfall runoff, and traffic behavior prediction. CA are generally described by a quaternion $a = f\ (L,\ s,\ n)$, where L represents the d-dimensional cell space, s represents the finite-state set of cell, and n represents the set of neighbor cells (including the central cell). As shown in Fig. 4, the common neighborhoods are mainly of the von Neuman or Moore types [40], where the central and neighbor cells are represented by the red and blue grid points, respectively.

The state of each cell at the next time step is determined by the state of this cell and its neighbor cells at the current time step. This state update can be expressed mathematically as

$$S_i^{t+1} = F\left(S_i^t, S_N^t\right) \tag{9}$$

where F represents the state-transition function, t is the current time step, S_i^t is the current state of the cell i, S_N^t is the current state of the neighbors of the cell i, and S_i^{t+1} is the state of the cell i at the next time step. The states of all cells are updated simultaneously in parallel. The transition function F is independent of the cell and time indices, and is a core component of CA. In general, different transition functions can be defined to meet the needs of different practical applications.

The hyperspectral CA method is proposed as a variant of the conventional CA method. This proposed method is mainly used to solve the problem of subpixel feature-type discrimination and maximize the correlation between the features. The differences between the hyperspectral and the conventional CA methods are as follows.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 04:01:16 UTC from IEEE Xplore. Restrictions apply.

1) For the conventional CA model, each cell is considered to be static, that is, only the state of each cell changes during evolution, while the cell location remains fixed. The opposite occurs in the hyperspectral variant, i.e., cells exchange positions, while their states remain unchanged during evolution.
2) In every evolution step of the conventional CA model, all cells change their states at the same time. For the hyperspectral variant, each evolution step involves a series of exchange actions.
3) First, before subpixel localization, it is necessary to perform edge detection and divide image pixels into edge and nonedge pixels. The nonedge pixels are the pixels of the same kind of a ground object in a subpixel. These pixels can directly copy the pixel features, while the edge pixels are used for subpixel localization. In a hyperspectral cellular automaton, the subpixel obtained by the average segmentation of the original low-resolution pixel is a cell. The cell space refers to the subpixel space obtained by an average pixel. The cell state corresponds to one of the N types of ground objects. The boundary condition is represented by other low-resolution pixels around the cell space, while the resulting subpixels are called boundaries. In this article, the used neighbor model is of Moore type [41], which is shown in Fig. 4(a).

The evolution steps for the hyperspectral cellular automaton model are as follows.

1) Initialization: The system assigns a corresponding state to all cells in the cell space, i.e., each cell corresponds to an object type. The cell state can be any of N different types, corresponding to the number of ground objects. However, the number of cells required to satisfy the same state is the same as the number M of subpixels occupied by the object in the low-resolution pixels. The probability of loss exchange m is a small number, which is herein set to 0.05.

2) Evolution: The system evolves, where each stage is determined by an n-step subprocess ($n = N \times N$), which can be described as follows.

1) Randomly select a cell, neither of whose neighbor states is the same as its own state.
2) Randomly select another cell in the cell space if the cell state is the same as the original cell state. If all the neighbor states of the cell are the same, then the subprocess is ended. Otherwise, go to Step 3.
3) Calculate the *exchange benefit* G (which represents the sum of the number of neighbor cells with the same state and the cell state after exchanging two cells minus the number of neighbor cells with the same state and the cell state before the exchange). A strictly positive exchange benefit, $G > 0$, means that exchanging two cells can improve the overall spatial correlation. Otherwise, the two cells can be exchanged with a loss probability m.
4) Termination of the subprocess. In actual calculations, all pixels of the original low-resolution HSI can be processed simultaneously in each step to improve the computational efficiency.

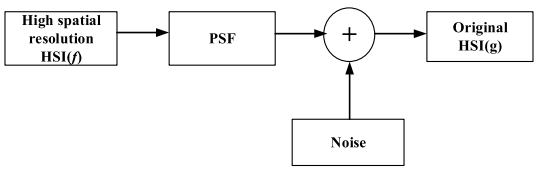


Fig. 5. HSI SR model.

C. SR Processing for HSIs

The SR processing for HSIs refers to the process of increasing the original spatial resolution of an image through the analysis and inversion of the image-degradation process. Through the abovementioned hyperspectral CA method, a preliminary superresolved image can be obtained, which is constructed using the spectral image information and subpixel mapping. Next, SR processing is carried out for the whole HSI through a variant of the method of projections onto convex sets (POCS) [42]. This method can be used in image SR to look for an effective solution set from a given initial result. The traditional POCS method has three main steps: 1) constructing a reference frame; 2) determining a constraint set; and 3) looping for the corrected image. During reference-frame construction, the traditional POCS algorithm usually interpolates the original image to get a high-resolution (HR) image. This process results in a lot of errors.

Apart from the traditional POCS method, the hyperspectral SR method feeds the preliminary superresolved image as the reference or the initial frame for an improved POCS algorithm. This hyperspectral SR method shows more suitability for the HSIs with a large numbers of bands, produces more accurate frames, fuses spectral information seamlessly into the spatial image distribution, and, hence, streamlines the subsequent reconstruction process.

An SR model associates a HSI of high spatial resolution and its corresponding low-spatial-resolution image, as shown in Fig. 5. Prior knowledge of the point-spread function (PSF) and the noise statistical characteristics in the imaging system can be regarded as the constraints on the image-reconstruction results. Each of these constraints corresponds to a convex set $C_k, k = 1, 2, \ldots, K$ with an ideal SR image [43].

The image SR model can be expressed as

$$g = H * f + n \tag{10}$$

where g represents the low-spatial-resolution image, H represents a point-spread function, f represents the high-spatial-resolution image, and N represents the noise.

In order to simplify the analysis and enhance the anomaly-detection performance, the point-spread function used in this article is a 2-D Gaussian function [44], which is expressed as

$$H(x, y) = e^{-[(x-x_1)^2+(y-y_1)^2]/2} \tag{11}$$

where (x_1, y_1) is an HSI pixel targeted for SR. As shown in Fig. 6, the POCS algorithm enforces every image pixel to satisfy all convex-set constraints. Suppose that there are T prior pieces of knowledge and K closed convex sets. Let P_j denote the projection operator of the jth closed convex set.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 04:01:16 UTC from IEEE Xplore. Restrictions apply.

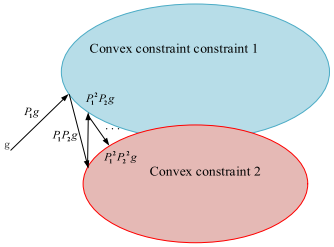


Fig. 6. Convex-set constraint diagram.

Then, the HR images are obtained as follows:

$$f_{i+1} = P_K P_{K-1}, \ldots, P_2 P_1\{f_i\} = P f_i. \tag{12}$$

The above convex sets can be defined as [45]

$$C_k(x, y) = \left\{ \hat{f}(x, y) : |r_k(x_1, y_1)| \le \delta_0 \right\} \tag{13}$$

$$r_k(x_1, y_1) = g_k(x_1, y_1) - \sum_{x=1}^{M-1} \sum_{y=1}^{N-1} \hat{f}(x, y) h_k(x, y, x_1, y_1) \tag{14}$$

where $g(x, y)$ and $f(x, y)$ represent the corresponding HSIs of the low and high spatial resolutions, respectively. The function $\hat{f}(x, y)$ represents an estimate of the ideal HR HSI, $h(x, y, x_1, y_1)$ is a PSF centered at the pixel (x_1, y_1), $\delta = c\sigma_0$ is the maximum allowable error that reflects the ideal high resolution, while σ_0 represents the noise variance of the low-spatial-resolution HSI. The degree of statistical confidence that the ideal high-spatial-resolution image $f(x, y)$ is located in C_k is obviously 99% when $c = 3$. The function $r(x, y)$ represents the residual of the true and observed image values. This residual function has the same statistical distribution as that of the noise. Therefore, the difference between the true and observed image values can be defined by the noise-confidence interval.

The projection of an HSI point $\hat{f}(x, y)$ in $C_k(x, y)$ should satisfy

$$P_k\left|\hat{f}(x, y)\right| = \hat{f}(x, y) + \begin{cases} \dfrac{r_k(x, y) - \delta_0}{\sum_x \sum_y h_k^2(x, y, x_1, y_1)} h(x, y, x_1, y_1), \\ \qquad r_k(x, y) > \delta_0 \\ 0, \qquad |r_k(x, y)| < \delta_0 \\ \dfrac{r_k(x, y) + \delta_0}{\sum_x \sum_y h_k^2(x, y, x_1, y_1)} h(x, y, x_1, y_1), \\ \qquad r_k(x, y) < -\delta_0 \end{cases} \tag{15}$$

where $P_k|(x, y)|$ is the estimation result after the constrained projection. In order to ensure the satisfaction of the convex-set constraints, the magnitude constraint C_A is increased as

$$C_A = \{f(i, j) : 0 \le f(i, j) \le 255\}. \tag{16}$$

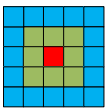


Fig. 7. Double-window structure.

The projection operator of the corresponding amplitude constraint is

$$C_A = \begin{cases} 0, & f(i, j) < 0 \\ f(i, j), & 0 \le f(i, j) \le 255 \\ 255, & f(i, j) > 255. \end{cases} \tag{17}$$

Therefore, the estimated ideal HR image $f(x, y)$ is projected as

$$f^{k+1}(x, y) = \prod_{(0,0) \le (x_1, y_1) \le (M-1, N-1)} P_k(x_1, y_1) \left| f^{k\wedge}(x, y) \right|. \tag{18}$$

From (18), the update process can be stopped when the error value reaches a certain threshold. Thereby, the high spatial resolution image is obtained by the hyperspectral SR method.

Now, we find the computational complexity of the SR method. For aHSI, denote by n, d, t, and m the number of pixels, the number of bands, the number of mixing pixels, and the number of ground objects, respectively. Therefore, the computational complexity of subpixel location and mapping is $O(4np)$, while the computational complexity of image SR in a single band is $O((n)^2 + 4\,mt + (n - t))$. Therefore, the computational complexity of SR processing for the whole HSI is $O(4np + d(n^2 + 4\,mt + (n - t)))$.

D. Collaborative Correlation for Anomaly Detection

Anomaly detection based on collaborative correlation follows mainly two steps. First, through feature correlation, the pixels that contain the anomalous parts in the background are removed provided that the residuals of the tested pixel and the background area are small. This step effectively reduces the false alarms. The second step is to enhance the target by subtracting the background area after anomaly removal and pixel testing by the GRX detector. Finally, the above two steps are combined, and with a given threshold, the anomaly-detection results can be obtained.

Fig. 7 shows the double-window structure, in which the red part represents the tested pixel, the green part represents the protecting pixels, and the blue part represents the surrounding pixels.

The ground-object correlation is reflected in the similarity among the test pixel, the background, and the anomaly pixels [46]. The feature correlation is based on the following assumptions. The background in the tested pixels can be expressed as atoms in the background dictionary, while the anomaly pixels cannot. The null and alternative hypotheses of anomaly detection using ground-object correlation can be expressed, respectively, as

$$M_0 : \mathbf{y} = \mathbf{A}\boldsymbol{\alpha} + \mathbf{n} \text{ (Anomaly absent)} \tag{19}$$

$$M_1 : \mathbf{y} \ne \mathbf{A}\boldsymbol{\alpha} + \mathbf{n} \text{ (Anomaly present)} \tag{20}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 04:01:16 UTC from IEEE Xplore. Restrictions apply.

where A represents the ground objects of the background dictionary for the blue area of the double-window structure and α represents the abundance value for A (also known as the dictionary atoms).

The endmember spectrum contained in a single pixel must satisfy the nonnegativity and the unit-sum conditions for the abundance values. Therefore, the α values representing the tested pixel also have similar conditions

$$\mathbf{e}^T\boldsymbol{\alpha} = 1 \tag{21}$$

$$\alpha_i \geq 0, \quad i = 1, \ldots, N. \tag{22}$$

In general, an endmember with a large abundance value is more likely to correspond to an anomaly, and the maximum value in the tested pixel is denoted by C. Then, the condition in (20) becomes

$$0 \leq \alpha_i \leq C. \tag{23}$$

According to (21), C might be viewed as the maximum value of the background abundance around the neighborhood of the pixel of interest, and the range of C is the unit interval, i.e., $C \in (0, 1)$

$$l(i) = \text{mean}(\|\mathbf{Y} - \mathbf{A}_i\boldsymbol{\alpha}_i\|_2), \quad i = 1, \ldots, N. \tag{24}$$

The array $\mathbf{Y}$ represents the protected pixels around the test pixel y. Moreover, when $l(i)$ reaches the maximum value, α_i is considered to be a potential anomaly target atom, and its corresponding dictionary A_i is removed from A. The final background dictionary and the corresponding atom are updated to A' and α'. The rest of the pixels around the background are used as the background for anomaly detection with the tested pixel.

The objective function in (25) represents the residual between the tested pixel and the surrounding background area, which can be obtained as

$$\mathbf{r} = \left\|\mathbf{y} - \mathbf{A}'\boldsymbol{\alpha}'\right\|_2. \tag{25}$$

Through (25), we can get the features and abundance values that make the residual larger in the surrounding background area. By removing these features, we can effectively ensure that the background area contains only the background with no anomaly parts. After that, the final anomaly detector can be obtained by combining the residual with the GRX detector.

The GRX-detection method with the double-window structure assumes that the background confirms to a multidimensional normal distribution when the target and the background are unknown. This method tests the null hypothesis H_0 that the target does not exist, against the alternative hypothesis H_1 that the target exists [47]

$$H_0 : \mathbf{y} = \mathbf{n} \text{ (Anomaly absent)} \tag{26}$$

$$H_1 : \mathbf{y} = \mathbf{as} + \mathbf{n} \text{ (Anomaly present)} \tag{27}$$

where n and s represent the background noise and the target spectral signal, respectively, and a is a signal coefficient. Based on the binary hypothesis (26) and (27), the detection criterion based on the constant false-alarm rate (CFAR) can be formulated as

$$\text{RX}(\mathbf{y}) = (\mathbf{y} - \mu)^T\Gamma(\mathbf{y} - \mu) \underset{<H_0}{\overset{\geq H_1}{}} \eta \tag{28}$$

Algorithm 1 The Proposed SRP-SC Algorithm for Anomaly Detection

Inputs: Hyperspectral image, Double-window structure (win_{out} win_{in}),

Steps:

1. Perform ELMM-based hyperspectral unmixing, and obtain the endmember spectrum with the parameters φ and the abundance maps of the ground objects.
2. Divide the pixels into four parts, as shown in Fig. 3. Perform subpixel localization and mapping of two or more objects by the cellular automata method.
3. Super-resolve the HSI using the hyperspectral POCS method.
4. For each pixel y,
   1) Setup the double-window structure with the window size centered at y.
   2) Remove the possible anomaly part of the neighboring pixels using (24).
   3) Calculate the collaborative-correlation-based detection function value R (30) and determine whether an anomaly target is present or not using the given threshold.
4. Calculate the detector output.

where y is the pixel to be detected in the inner window, μ is the average value of the background pixel in the outer window, Γ is the covariance matrix of the background pixels in the outer window, and η is the decision threshold. The RX algorithm obtains a point-by-point detection in the HSI according to the criterion in (28).

Finally, the overall detection result is obtained by combining the residual and RX criteria of (25) and (28)

$$R = \left\|\mathbf{y} - \mathbf{A}'\boldsymbol{\alpha}'\right\|_2 + (\mathbf{y} - \mu)^T\Gamma(\mathbf{y} - \mu). \tag{29}$$

If R is larger than a given threshold, y is labeled as an anomaly pixel. The complete anomaly-detection procedure is summarized in Algorithm 1.

III. Experimental Results

A. Hyperspectral Data Set Description

In this section, the proposed method for anomaly detection is applied to six hyperspectral data sets to verify the effectiveness of the method. These sets are described as follows.

1) *Airport-Beach-Urban Database:* The Airport-Beach-Urban (ABU) database is publically available, and its detailed acquisition procedure is described in [48]. The data set images have 100×100 pixels or 150×150 pixels with the corresponding references. The airplane and urban scenes, as well as most of the beach scenes, were captured by an airborne visible/infrared imaging spectrometer (AVIRIS) sensor. The symbols HSI-I, HSI-II, and HSI-III are used to denote these three kinds of HSIs.
2) *San Diego Data Set:* This real HSI data set is widely used and is also collected by an AVIRIS sensor. The scene is a part of the San Diego airport area, CA, USA [23]. The HSI data contain 100×100 pixels in each band where the noisy bands have been removed.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 04:01:16 UTC from IEEE Xplore. Restrictions apply.

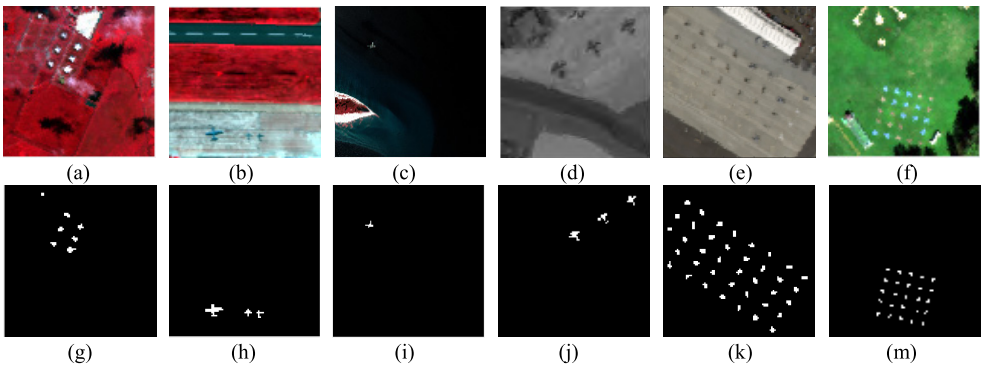


Fig. 8. HSI data set description. (a) False-color image of HSI-I. (b) False-color image of HSI-II. (c) False-color image of HSI-III. (d) False-color image of HSI-IV. (e) False-color image of HSI-V. (f) False-color image of HSI-V. (g) Ground truth of HSI-I. (h) Ground truth of HSI-II. (i) Ground truth of HSI-III. (j) Ground truth of HSI-IV. (k) Ground truth of HSI-V. (m) Ground truth of HSI-V.

TABLE I
DETAILED INFORMATION OF SIX HSIS

HSI	Size	Capturing location	Spatial resolution	Sensor
HSI-I	100×100	Texas Coast	17.2m	AVIRIS
HSI -II	100×100	Gulfport	3.4m	AVIRIS
HSI -III	150×150	Cat Island	17.2m	AVIRIS
HSI -IV	100×100	San Diego	7.1m	AVIRIS
HSI -V	100×100	San Diego	3.5m	AVIRIS
HSI -VI	80×80	Rochester	1m	ProSpec

The sets HSI-IV and HSI-V constitute specific sets of this data set.

3) *Avon Data Set:* The last data set (HSI-V) is the Avon data set, which was collected using a push-broom hyperspectral ProSpec TIR-VS sensor during the SpecTIR Hyperspectral Airborne Experiment 2012 (SHARE 2012) data-collection campaign [49]. From the morning reflectance data, a region of 80×80 pixels was selected for the experiments. This region covers a driving park in Avon, south of Rochester, New York, USA. This data set has 360 bands covering a spectral range of 400–2450 nm, with a spectral resolution of 5 nm. There are 25 tarps and three red or blue felts with 68 target pixels to be detected in a scene. These HSIs contain anomaly targets and corresponding reference maps, as shown in Fig. 8. The detailed information of the six data sets is shown in Table I.

B. Compared Methods and Evaluation Criteria

To verify the detection performance and robustness of the proposed algorithm, the GRX, local-RX, CRD, abundance- and dictionary-based low-rank (ADLR), CNND, and low-rank and sparse matrix decomposition (LRaSMD) detectors are compared.

The RX algorithm [13] is the most commonly used hyperspectral anomaly-detection method. This method detects the anomalies through the comparison of the difference between the mean square of the tested and background pixels. According to the range of the calculated mean and variance, the RX method can be divided into GRX and local-RX methods. A GRX method calculates the mean and variance based on entire HSIs, while a local-RX method calculates the mean and variance locally for the outer window of a double-window structure.

The collaborative-representation detector (CRD) method [22] was developed based on the fact that the background can be well represented by its spatial surroundings, while the anomaly targets cannot. Collaboration is achieved through a linear combination of the surrounding neighbors of each test sample. The optimal weight vector in a double-window structure is obtained through l_2-norm minimization. For the experiments in this work, the regularization parameter was set to 10–5, following [22].

The ADLR-decomposition method [26] obtains the endmember types and abundance maps of various features in the HSI through unmixing and establishes the background dictionary and anomaly target dictionaries according to the abundance of various features. Thus, a decomposition model of the HSIs is established, using the alternating direction method of multipliers (ADMM). The model is exploited to obtain the anomaly-detection results.

The convolutional neural network detector (CNND) method [20] performs CNN network training and testing using HSIs captured by the same sensor. Through the pixel pair training method, a network is obtained for anomaly detection, and testing is performed for the double-window structure. Specifically, each pixel and its surrounding pixels form a pixel pair that is fed into the trained network to get an abnormality evaluation score, which is the thresholded to achieve anomaly detection.

The LRaSMD method [23] decomposes the HSI into background and anomaly parts. Thus, the background part can be represented by a dictionary multiplying the corresponding coefficient matrix. In LRaSMD, the regularization parameters are set as $\lambda = 0.01$ and $\beta = 0.1$.

The area under the curve (AUC) of the receiver operating characteristics (ROC) [50] is used for quantitative performance evaluation, and is the most widely used metric in hyperspectral anomaly detection. The true-positive rate (TPR) is the number of correct positive detections occurring among all positive

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 04:01:16 UTC from IEEE Xplore. Restrictions apply.

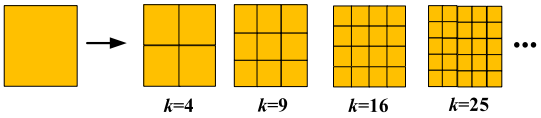


Fig. 9. Diagram of the single pixel divided into subpixels.

TABLE II
TIME-BASED ANOMALY-DETECTION PERFORMANCE FOR DIFFERENT NUMBER OF PIXEL SEGMENTATIONS

k	4	9	16	25
HSI-I	128.44	**148.25**	201.92	289.15
HSI-II	132.25	**185.72**	225.73	302.42
HSI-III	118.43	**168.47**	221.64	296.27
HSI-IV	135.34	**155.86**	203.48	312.44
HSI-V	143.28	**177.63**	224.96	305.38
HSI-V	95.26	**152.37**	215.74	300.86

TABLE III
AUC-BASED ANOMALY-DETECTION PERFORMANCE FOR DIFFERENT NUMBER OF PIXEL SEGMENTATIONS

k	4	9	16	25
HSI-I	0.9985	**0.9432**	0.9119	0.8534
HSI-II	0.9923	**0.9556**	0.9248	0.8207
HSI-III	0.9976	**0.9423**	0.9127	0.8178
HSI-IV	0.9943	**0.9678**	0.9271	0.7963
HSI-V	0.9877	**0.9321**	0.9076	0.7742
HSI-V	0.9948	**0.9268**	0.8952	0.8362

TABLE IV
AUC-BASED ANOMALY-DETECTION PERFORMANCE FOR THE HSI-I DATA SET AND DIFFERENT WINDOW SIZES

	(3,5)	(5,7)	(7,9)	(7,13)
SRP-SC	0.9912	**0.9985**	0.9935	0.9834
LRX	0.9867	**0.9961**	0.9893	0.9786
CRD	0.9824	**0.9922**	0.9867	0.9748
CNND	0.9764	**0.9943**	0.9794	0.9725

TABLE V
AUC-BASED ANOMALY-DETECTION PERFORMANCE FOR THE HSI-II DATA SET AND DIFFERENT WINDOW SIZES

	(3,5)	(5,7)	(7,9)	(7,13)
SRP-SC	**0.9923**	0.9901	0.9897	0.9825
GRX	**0.8735**	0.8721	0.8635	0.8612
CRD	**0.9833**	0.9826	0.9786	0.9635
CNND	**0.8923**	0.8893	0.8816	0.8617

TABLE VI
AUC-BASED ANOMALY-DETECTION PERFORMANCE FOR THE HSI-III DATA SET AND DIFFERENT WINDOW SIZES

	(3,5)	(5,7)	(7,9)	(7,13)
SRP-SC	**0.9976**	0.9927	0.9894	0.9875
LRX	**0.9943**	0.9865	0.9812	0.9801
CRD	**0.9813**	0.9792	0.9765	0.9724
CNND	**0.9768**	0.9635	0.9621	0.9605

TABLE VII
AUC-BASED ANOMALY-DETECTION PERFORMANCE FOR THE HSI-III DATA SET AND DIFFERENT WINDOW SIZES

	(3,5)	(5,7)	(7,9)	(7,13)
SRP-SC	**0.9976**	0.9927	0.9894	0.9875
LRX	**0.9943**	0.9865	0.9812	0.9801
CRD	**0.9813**	0.9792	0.9765	0.9724
CNND	**0.9768**	0.9635	0.9621	0.9605

TABLE VIII
AUC-BASED ANOMALY-DETECTION PERFORMANCE FOR THE HSI-IV DATA SET AND DIFFERENT WINDOW SIZES

	(3,5)	(5,7)	(7,9)	(7,13)
SRP-SC	0.9635	0.9867	0.9912	**0.9943**
LRX	0.6254	0.6873	0.6901	**0.6914**
CRD	0.9134	0.9384	0.9487	**0.9530**
CNND	0.9328	0.9562	0.9586	**0.9652**

TABLE IX
AUC-BASED ANOMALY-DETECTION PERFORMANCE FOR THE HSI-VI DATA SET AND DIFFERENT WINDOW SIZES

	(3,5)	(5,7)	(7,9)	(7,13)
SRP-SC	**0.9948**	0.9915	0.9886	0.9837
LRX	**0.9205**	0.9168	0.9081	0.8635
CRD	**0.9605**	0.9562	0.9435	0.9264
CNND	**0.9723**	0.9674	0.9582	0.9281

samples, while the false-positive rate (FPR) is the number of incorrect positive detections occurring among all negative samples [51]. The ROC curve can describe the anomaly-detection performance with the TPR plotted as a function of FPR at each threshold. The AUC gives a quantitative summary of the ROC curve.

C. Parameter Analysis

In this section, we investigate the impacts of the parameters such as unmixing processing, SR processing, and detection in the proposed method, and choose the best parameter to carry on the comparative experiment.

1) Unmixing Process: There are two parameters in the unmixing process: the spectral parameter ψ of spectral variability in ELMM and different unmixing methods. The parameter in ELMM is the endmember spectral parameter ψ. In the unmixing process, we consider all kinds of terrain unmixing component maps obtained by ELMM. These maps are used in the SR process to obtain accurate superresolved images. The parameter ψ is used in the unmixing process to reflect accurately the spectral changes of the anomaly targets. Therefore, the proposed method does not consider the influence of this parameter on the anomaly-detection results, and it considers the influence on other unmixing methods that do not consider the spectral changes. This discussion is carried out in detail in the Section III-E.

2) SR Process: The SR process includes subpixel localization and mapping and SR processing for the HSIs. The parameter in the SR process is the number k of pixel segmentation.

Fig. 9 shows the diagram of the single pixel divided into subpixels, and Tables II and III show the influence of different k values on the calculation time and AUC value. It can be seen that when k is 4, the detection result is optimal and the minimum calculation time consumption is achieved.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 04:01:16 UTC from IEEE Xplore. Restrictions apply.

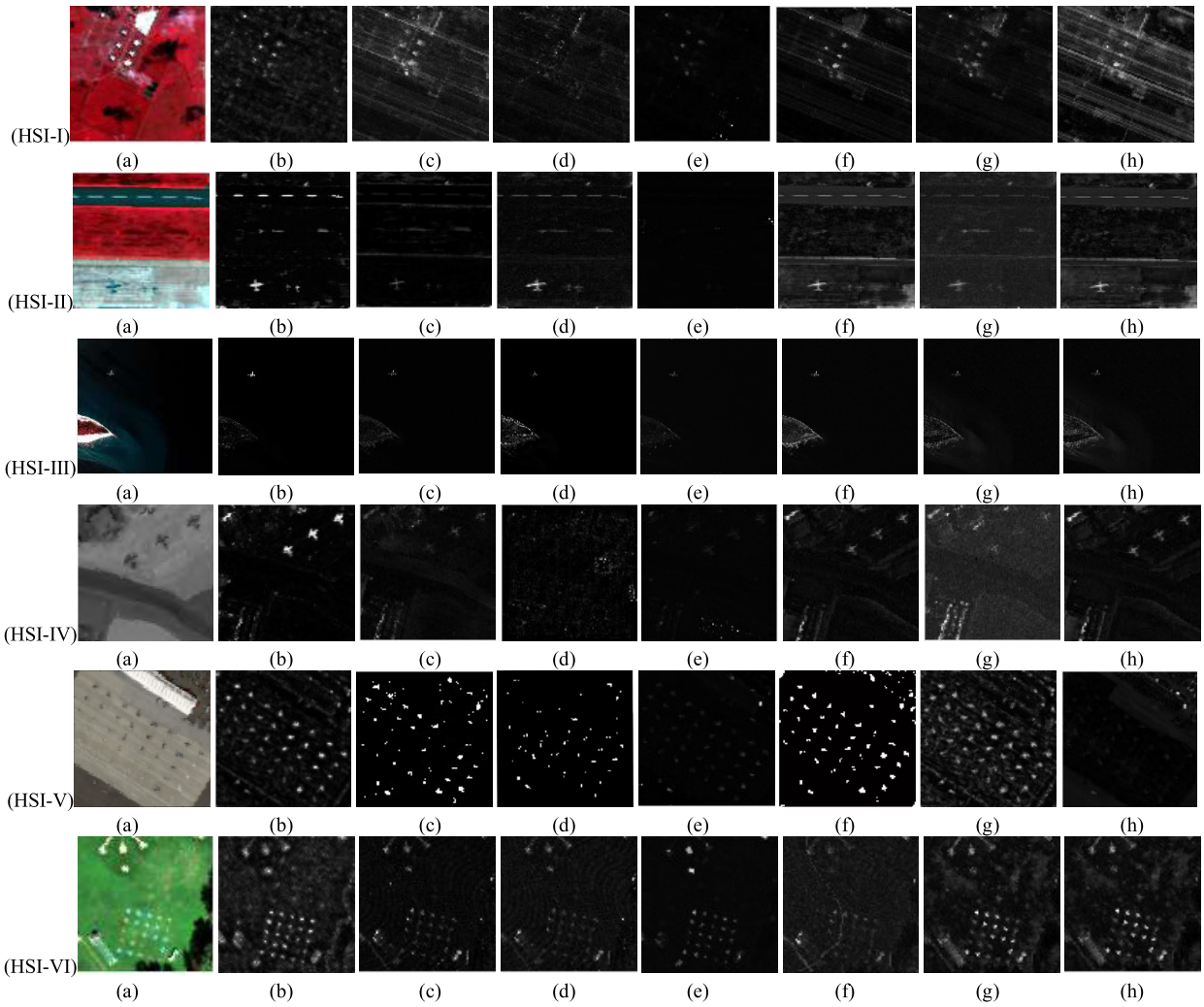


Fig. 10. Anomaly-detection performance results for the compared methods and six HSI data sets. (a) Input image scene. (b) Proposed method. (c) Local RX. (d) GRX. (e) CRD. (f) ADLR. (g) CNND. (h) LRaSMD.

When k is increased, the AUC value is increased gradually. This is because when the number of subpixels is increased, the time of the SR process is increased, errors are introduced in this process, and the spatial correlation and ground-object correlation are destroyed, resulting in the decrease in the AUC value. Therefore, the value of parameter k in the SR process is 4.

3) Detection Process: The main parameter in the detection process is the window-size parameter settings. The proposed method [super-resolution processing and spatial correlation (SRP-SC)] as well as the GRX, CRD, and CNND methods have window-size parameter settings that are affected by the anomaly target size. In this section, the optimal window size is experimentally selected. Tables IV–IX compare the anomaly-detection performance of the above methods based on the AUC metric.

From Tables IV–IX, it can be seen that the AUC values of our SRP-SC method are higher than the corresponding values of the other methods on different HSI data sets. This indicates that the proposed method can effectively enhance the target and suppress the background, and hence, this method can be useful in detecting and tracking the anomaly targets. Moreover, from these tables, the best window-size settings of the proposed method and the compared methods for the HSI-I and HSI-IV data sets are (5, 7) and (7, 13), respectively. For all other data sets, the best window size is (3, 5).

D. Detection Performance

In this section, the detection performance of the proposed method (SRP-SC) is evaluated and compared against the five other detectors: GRX, Local-RX, CRD, ADLR, CNND, and LRaSMD. With the best window-size parameter settings, the anomaly-detection results are visually shown in Fig. 10. From these results, the SRP-SC method can detect different scenes including the airport, beach, and urban scenes. Another advantage of the SRP-SC method is that it can detect anomaly targets with different sizes, especially small targets. The SRP-SC method has better performance on fine details than the other detectors, and it can detect small targets with several pixels or subpixels clearly.

Fig. 11 shows the ROC curves of the six compared detectors on the six selected data sets. Obviously, the proposed SRP-SC method always shows higher ROC curves and superior performance compared with the LRX, GRX, CRD, ADLR, CNND,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 04:01:16 UTC from IEEE Xplore. Restrictions apply.

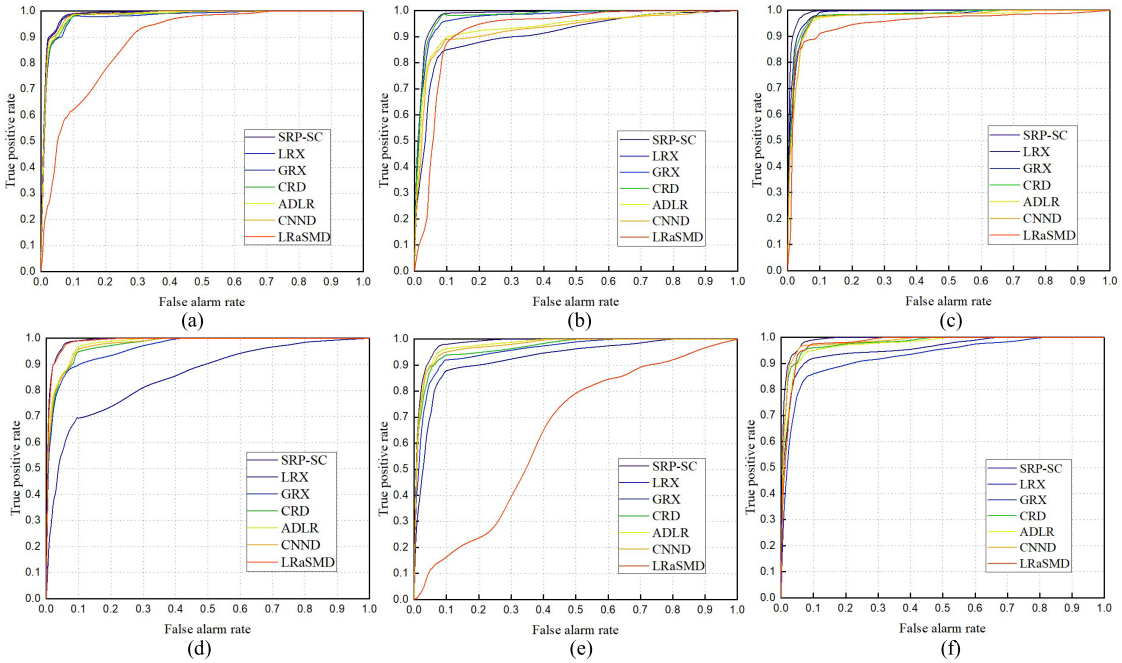


Fig. 11. ROC curves of different anomaly-detection methods for different HSI data sets. (a) HSI-I. (b) HSI-II. (c) HSI-III. (d) HSI-IV. (e) HSI-V. (f) HSI-VI.

TABLE X
AUC-BASED ANOMALY-DETECTION PERFORMANCE ON DIFFERENT SIX DATA SET WITH DIFFERENT METHODS AND OPTIMAL WINDOW SIZE

HSIs	SRP-SC	LRX	GRX	CRD	ADLR	CNND	LRaSMD
HSI-I	**0.9985**	0.9961	0.9904	0.9922	0.9931	0.9943	0.9501
HSI-II	**0.9923**	0.8735	0.9526	0.9833	0.9023	0.8923	0.9532
HSI-III	**0.9976**	0.9943	0.9805	0.9813	0.9723	0.9768	0.9189
HSI-IV	**0.9943**	0.6914	0.9085	0.9530	0.9772	0.9652	0.9907
HSI-V	**0.9877**	0.8870	0.9231	0.9406	0.9602	0.9503	0.6363
HSI-VI	**0.9948**	0.9205	0.8625	0.9605	0.9543	0.9723	0.9742

TABLE XI
FALSE-ALARM RATES ON SIX DATA SETS WITH DIFFERENT METHODS AND THE OPTIMAL WINDOW SIZE

HSIs	SRP-SC	LRX	GRX	CRD	ADLR	CNND	LRaSMD
HSI-I	0.0245	0.0372	0.0627	0.0563	0.0486	0.0424	0.2799
HSI-II	0.0369	0.3005	0.0538	0.0474	0.1012	0.2015	0.1206
HSI-III	0.0164	0.0319	0.0460	0.0396	0.0524	0.0474	0.0934
HSI-IV	0.0215	0.5009	0.1051	0.0704	0.0679	0.0758	0.0342
HSI-V	0.0383	0.2015	0.0832	0.0601	0.0510	0.0576	0.7425
HSI-VI	0.0255	0.0718	0.2170	0.0461	0.0549	0.0294	0.0468

and LRaSMD methods. This observation confirms that the proposed method can be (and should be) applied to anomaly detection in different scenarios. And a similar conclusion can be drawn from Table X.

Table XI shows the false-alarm rate of each method when the anomaly-detection rate reaches 0.9 for different HSIs. As shown in Table XI, when the detection rate of the proposed method reaches 0.9, the false-alarm rate is the lowest compared with other methods. This shows that the proposed method can effectively suppress the background. From the experimental results, we can see that the proposed method can effectively protect the details of small anomalous targets, and hence achieve accurate anomaly detection. In addition, the exploitation of two spatial correlation types, namely, the pixel correlation and the ground-object correlation, clearly boosts the detection performance. Indeed, the combination of the two correlation types can effectively remove the false alarms, and consequently enhance the anomaly detection of the hidden targets.

The computation time of the proposed method is also compared with those of the competing methods. All experiments are carried out using MATLAB on a machine with an Intel Core i7-7700 CPU and 8-GB RAM. The computation time (in seconds) for the experimental data is listed in Table XII. It is obvious that global methods have less computation times than the local ones. Although the proposed method takes less time than the CNND, it takes more computation time than the other methods. This is because the proposed method takes time in SR processing. The CNND method uses the whole scene to train the CNN network, incurs an enormous computational burden, and consumes time the most.

E. Comparison of Unmixing Methods

In this section, we analyze the effects of other hyperspectral unmixing methods on the detection performance. In particular, we compare the ELMM-based SVSC method as well as two other unmixing methods on the selected HSI data sets.

The minimum-volume-constrained nonnegative matrix factorization (MVC-NMF) algorithm [52] is an endmember-extraction algorithm based on convex geometry optimization. Under the assumption of the existence of pure pixels, this algorithm improves the precision of the endmember extraction and unmixing by adding the volume constraints of the convex

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 04:01:16 UTC from IEEE Xplore. Restrictions apply.

TABLE XII

COMPUTATION TIMES ON SIX DATA SETS WITH DIFFERENT METHODS AND THE OPTIMAL WINDOW SIZE

HSIs	SRP-SC	LRX	GRX	CRD	ADLR	CNND	LRaSMD
HSI-I	128.44	55.67	0.21	70.24	113.48	145.69	36.30
HSI-II	132.25	58.31	0.37	73.25	124.91	167.32	31.75
HSI-III	118.43	49.35	0.18	69.35	107.33	138.20	31.64
HSI-IV	135.34	35.18	0.27	53.85	107.85	182.27	28.06
HSI-V	143.28	39.42	0.36	67.48	110.27	204.36	35.83
HSI-VI	95.26	27.38	0.42	35.86	89.42	112.41	26.82

TABLE XIII

RMSE-BASED UNMIXING PERFORMANCE FOR DIFFERENT METHODS WITH THE OPTIMAL WINDOW SIZE AND HSI DATA SETS

Method	HSI-I	HSI-II	HSI-III	HSI-IV	HSI-V	HSI-VI
ELMM	10.22	4.43	6.87	8.68	11.25	13.23
MVC-NMF	16.28	12.35	12.56	12.35	18.24	17.84
R-Co NMF	14.35	9.81	8.76	16.67	16.21	15.27

cones and the single-side constraints to the conventional NMF algorithm. A sparseness constraint is applied on both the endmembers and the abundance matrices in each layer. In this way, the given data matrix can be sparsely decomposed even when it is not sparse. Robust collaborative NMF (R-CoNMF) [53] solves the problem by removing the redundant endmembers with collaborative regularization.

For real data, the most commonly used quantitative evaluation indicator of the unmixing methods is the root-mean-square error (RMSE) [54]. The RMSE of a HSI can be calculated as

$$\mathrm{RMSE}(l) = \sqrt{\frac{1}{N}\sum_{k=1}^{N}\left\|X_{kl} - \hat{X}_{kl}\right\|_2^2} \tag{30}$$

where l represents the lth band, N represents the number of pixels, while X indicates the original HSI, and $\hat{X}$ is the reconstructed HSI. The overall RMSE is the mean value over all HSI bands.

Table XIII shows the RMSE values on the different HSI data sets by the ELMM method and the other unmixing methods. Clearly, the ELMM method gives the best results, effectively protects the anomalous target information, improves the accuracy for the next subpixel localization, and reduces the detection errors.

Fig. 12 shows the influence of different unmixing methods on the anomaly-detection performance. After the MVC-MNF and R-CONMF unmixing methods are used to obtain the abundance maps, the AUC values of the anomaly detections corresponding to each hyperspectral data set are obtained. Obviously, the ELMM method can consider spectral variability, as this method effectively reflects the spectral changes of the anomalous targets in the spatial neighborhood, protects the characteristics of such targets in the SR process, and obtains better anomaly-detection results. The higher the accuracy of the abundance map, the smaller the RMSE value. As well, the higher the accuracy of the unmixing, the higher the detection AUC value. As shown in Fig. 12, compared with the other two methods, the ELMM-detection results are the best.

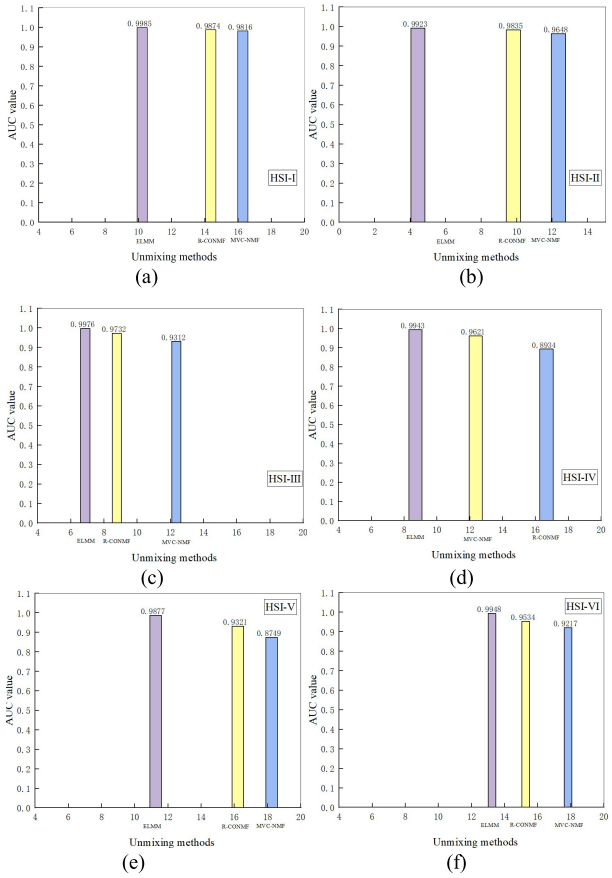


Fig. 12. Influence of different unmixing methods on the anomaly-detection performance. (a) HSI-I. (b) HSI-II. (c) HSI-III. (d) HSI-IV. (e) HSI-V. (f) HSI-VI.

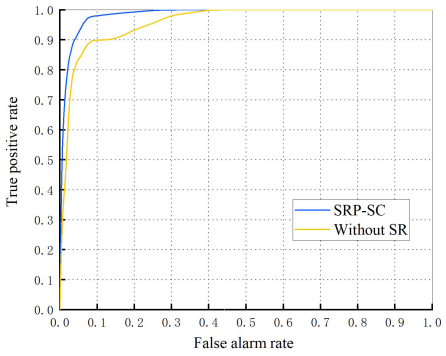


Fig. 13. ROC curves of the SRP-SC method with and without SR.

In addition, the proposed method is evaluated with and without SR on the HSI-V data set. The corresponding ROC curves are shown in Fig. 13. The AUC value of the SRP-SC method with SR is 0.9877, while the AUC value of the SRP-SC method without SR is 0.9326. From the results, SR can indeed enhance the detection performance by improving the HSI spatial resolution.

IV. CONCLUSION

In this article, a novel hyperspectral anomaly-detection algorithm based on SR and spatial correlation is proposed. To utilize fully the ground-object information and consider

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 04:01:16 UTC from IEEE Xplore. Restrictions apply.

the influence of the spectral variability in the mixed pixels on anomaly detection, an ELMM is applied with SR. This approach handles the spectral variability of the ground objects and improves the spatial resolution of the HSIs. The proposed algorithm makes better use of the rich HSI information and can avoid the influence of the mixed pixels on the anomalous targets. Subsequently, HSIs of higher spatial resolution and more ground-object details are used along with spatial correlation techniques for improved anomaly detection. The proposed algorithm can effectively suppress the background, highlight the anomalous targets, and obtain better detection performance. This article also provides a new SR scheme for the HSIs, which is helpful for the intelligent interpretation and application of the HSIs.

REFERENCES

[1] A. Plaza *et al.*, "Recent advances in techniques for hyper-spectral image processing," *Remote Sens. Environ.*, vol. 113, pp. S110–S122, Sep. 2009.

[2] D. Landgrebe, "Hyperspectral image data analysis," *IEEE Signal Process. Mag.*, vol. 19, no. 1, pp. 17–28, Jan. 2002.

[3] G. Cheng and J. Han, "A survey on object detection in optical remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 117, pp. 11–28, Jul. 2016.

[4] Y. Yuan, Q. Wang, and G. Zhu, "Fast hyperspectral anomaly detection via high-order 2-D crossing filter," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 2, pp. 620–630, Feb. 2015.

[5] S. Khazai, A. Safari, B. Mojaradi, and S. Homayouni, "An approach for subpixel anomaly detection in hyperspectral images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 6, no. 2, pp. 769–778, Apr. 2013.

[6] N. M. Nasrabadi, "Hyperspectral target detection: An overview of current and future challenges," *IEEE Signal Process. Mag.*, vol. 31, no. 1, pp. 34–44, Jan. 2014.

[7] M. Borengasser, W. S. Hungate, and R. Watkins, "Hyperspectral remote sensing: Principles and applications," *Iasri*, vol. 31, no. 12, pp. 1249–1259, Mar. 2007.

[8] C.-I. Chang and S.-S. Chiang, "Anomaly detection and classification for hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 40, no. 6, pp. 1314–1325, Jun. 2002.

[9] J. E. Fowler and Q. Du, "Anomaly detection and reconstruction from random projections," *IEEE Trans. Image Process.*, vol. 21, no. 1, pp. 184–195, Jan. 2012.

[10] A. Zare, C. Jiao, and T. Glenn, "Discriminative multiple instance hyperspectral target characterization," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 40, no. 10, pp. 2342–2354, Oct. 2018.

[11] S. Yang and Z. Shi, "Hyperspectral image target detection improvement based on total variation," *IEEE Trans. Image Process.*, vol. 25, no. 5, pp. 2249–2258, May 2016.

[12] Y. Yuan, D. Ma, and Q. Wang, "Hyperspectral anomaly detection by graph pixel selection," *IEEE Trans. Cybern.*, vol. 46, no. 12, pp. 3123–3134, Dec. 2016.

[13] I. S. Reed and X. Yu, "Adaptive multiple-band CFAR detection of an optical pattern with unknown spectral distribution," *IEEE Trans. Acoust., Speech, Signal Process.*, vol. 38, no. 10, pp. 1760–1770, Oct. 1990.

[14] R. Zhao, B. Du, and L. Zhang, "A robust nonlinear hyperspectral anomaly detection approach," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 4, pp. 1227–1234, Apr. 2014.

[15] H. Kwon and N. M. Nasrabadi, "Kernel RX-algorithm: A nonlinear anomaly detector for hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 2, pp. 388–397, Feb. 2005.

[16] A. P. Schaum, "Hyperspectral anomaly detection beyond RX," *Proc. SPIE*, vol. 6565, May 2007, Art. no. 656502, doi: 10.1117/12.718789.

[17] Q. Guo, B. Zhang, Q. Ran, L. Gao, J. Li, and A. Plaza, "Weighted-RXD and linear filter-based RXD: Improving background statistics estimation for anomaly detection in hyperspectral imagery," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 2351–2366, Jun. 2014.

[18] J. P. Williams, T. J. Bihl, and K. W. Bauer, "Towards the mitigation of correlation effects in anomaly detection for hyperspectral imagery," *J. Defense Model. Simul., Appl., Methodol., Technol.*, vol. 10, no. 3, pp. 263–273, Feb. 2013.

[19] S. Matteoli, M. Diani, and G. Corsini, "Improved estimation of local background covariance matrix for anomaly detection in hyperspectral images," *Opt. Eng.*, vol. 49, no. 4, p. 258, Apr. 2010.

[20] W. Li, G. Wu, and Q. Du, "Transferred deep learning for anomaly detection in hyperspectral imagery," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 5, pp. 597–601, May 2017.

[21] J. Li, H. Zhang, L. Zhang, and L. Ma, "Hyperspectral anomaly detection by the use of background joint sparse representation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 6, pp. 2523–2533, Jun. 2015.

[22] W. Li and Q. Du, "Collaborative representation for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 3, pp. 1463–1474, Mar. 2015.

[23] Y. Xu, Z. Wu, J. Li, A. Plaza, and Z. Wei, "Anomaly detection in hyperspectral images based on low-rank and sparse representation," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 4, pp. 1990–2000, Apr. 2016.

[24] W. Sun, C. Liu, J. Li, Y. M. Lai, and W. Li, "Low-rank and sparse matrix decomposition-based anomaly detection for hyperspectral imagery," *J. Appl. Remote Sens.*, vol. 8, no. 1, May 2014, Art. no. 083641.

[25] N. Huyan, X. Zhang, H. Zhou, and L. Jiao, "Hyperspectral anomaly detection via background and potential anomaly dictionaries construction," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 4, pp. 2263–2276, Apr. 2019.

[26] Y. Qu *et al.*, "Hyperspectral anomaly detection through spectral unmixing and dictionary-based low-rank decomposition," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 8, pp. 4391–4405, Aug. 2018.

[27] S. Zhang, J. Li, Z. Wu, and A. Plaza, "Spatial discontinuity-weighted sparse unmixing of hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 10, pp. 5767–5779, Oct. 2018.

[28] A. Plaza, Q. Du, J. M. Bioucas-Dias, X. Jia, and F. A. Kruse, "Foreword to the special issue on spectral unmixing of remotely sensed data," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 11, pp. 4103–4110, Nov. 2011.

[29] W. Xie, X. Jia, Y. Li, and J. Lei, "Hyperspectral image super-resolution using deep feature matrix factorization," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 8, pp. 4391–4405, Aug. 2018.

[30] R. Heylen, M. Parente, and P. Gader, "A review of nonlinear hyperspectral unmixing methods," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 1844–1868, Jun. 2014.

[31] C. Zhao and L. Zhang, "Spectral-spatial stacked autoencoders based on low-rank and sparse matrix decomposition for hyperspectral anomaly detection," *Infr. Phys. Technol.*, vol. 92, pp. 166–176, Aug. 2018.

[32] B. Somers, G. P. Asner, L. Tits, and P. Coppin, "Endmember variability in spectral mixture analysis: A review," *Remote Sens. Environ.*, vol. 115, no. 7, pp. 1603–1616, Jul. 2011.

[33] J. M. Molero, E. M. Garzon, I. Garcia, and A. Plaza, "Analysis and optimizations of global and local versions of the RX algorithm for anomaly detection in hyperspectral data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 6, no. 2, pp. 801–814, Apr. 2013.

[34] L. Drumetz, M.-A. Veganzones, S. Henrot, R. Phlypo, J. Chanussot, and C. Jutten, "Blind hyperspectral unmixing using an extended linear mixing model to address spectral variability," *IEEE Trans. Image Process.*, vol. 25, no. 8, pp. 3890–3905, Aug. 2016.

[35] H. Kim and H. Park, "Nonnegative matrix factorization based on alternating nonnegativity constrained least squares and active set method," *SIAM J. Matrix Anal. Appl.*, vol. 30, no. 2, pp. 713–730, Jan. 2008.

[36] W. Xie, X. Jia, Y. Li, and J. Lei, "Hyperspectral image super-resolution using deep feature matrix factorization," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 8, pp. 6055–6067, Aug. 2019.

[37] S. Zhang, A. Agathos, and J. Li, "Robust minimum volume simplex analysis for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 11, pp. 6431–6439, Nov. 2017.

[38] S. Gaur, K. M. Buddhiraju, and A. Porwal, "Sub-pixel mapping with hyperspectral images using super-resolution," in *Proc. IGARSS*, Jul. 2018, pp. 2459–2462.

[39] X. Xu *et al.*, "A new spectral-spatial sub-pixel mapping model for remotely sensed hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 8, pp. 4391–4405, Aug. 2018.

[40] V. A. Mardiris, G. C. Sirakoulis, and I. G. Karafyllidis, "Automated design architecture for 1-D cellular automata using quantum cellular automata," *IEEE Trans. Comput.*, vol. 64, no. 9, pp. 2476–2489, Sep. 2015.

[41] S. Nichele, M. B. Ose, S. Risi, and G. Tufte, "CA-NEAT: Evolved compositional pattern producing networks for cellular automata morphogenesis and replication," *IEEE Trans. Cogn. Devel. Syst.*, vol. 10, no. 3, pp. 687–700, Sep. 2018.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 04:01:16 UTC from IEEE Xplore. Restrictions apply.

[42] J. Huang, B. Gao, Y. Chen, Y. Chen, and L. Tong, "Super-resolution reconstruction of multi-polarization SAR images based on projections onto convex sets algorithm," in *Proc. IGARSS*, Jul. 2018, pp. 8456–8459.
[43] B. Wang, X. Chen, J. Li, and J. Cao, "An improved weighted projection onto convex sets method for seismic data interpolation and denoising," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 9, no. 1, pp. 228–235, Jan. 2016.
[44] L. Wang and P. Wang, "Sub-pixel mapping for hyperspectral imagery using super-resolution then spectral unmixing," in *Proc. IGARSS*, Jul. 2016, pp. 461–464.
[45] R. Heylen, M. A. Akhter, and P. Scheunders, "On using projection onto convex sets for solving the hyperspectral unmixing problem," *IEEE Geosci. Remote Sens. Lett.*, vol. 10, no. 6, pp. 1522–1526, Nov. 2013.
[46] Q. Ling, Y. Guo, Z. Lin, and W. An, "A constrained sparse representation model for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 4, pp. 2358–2371, Apr. 2019.
[47] X. Zhang, G. Wen, and W. Dai, "A tensor decomposition-based anomaly detection algorithm for hyperspectral image," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 10, pp. 5801–5820, Oct. 2016.
[48] X. Kang, X. Zhang, S. Li, K. Li, J. Li, and J. A. Benediktsson, "Hyperspectral anomaly detection with attribute and edge-preserving filters," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 10, pp. 5600–5611, Oct. 2017.
[49] A. M. Giannandrea *et al.*, "The SHARE 2012 data campaign," *Proc. SPIE*, vol. 8743, pp. 87430F-1–87430F-15, May 2013.
[50] C. Ferri, J. Hernández-Orallo, and P. Flach, "A coherent interpretation of AUC as a measure of aggregated classification performance," in *Proc. Int. Conf. Mach. Learn.*, 2011, pp. 657–664.
[51] D. Manolakis, E. Truslow, M. Pieper, T. Cooley, and M. Brueggeman, "Detection algorithms in hyperspectral imaging systems: An overview of practical algorithms," *IEEE Signal Process. Mag.*, vol. 31, no. 1, pp. 24–33, Jan. 2014.
[52] L. Miao and H. Qi, "Endmember extraction from highly mixed data using minimum volume constrained nonnegative matrix factorization," *IEEE Trans. Geosci. Remote Sens.*, vol. 45, no. 3, pp. 765–777, Mar. 2007.
[53] J. Li, J. M. Bioucas-Dias, A. Plaza, and L. Liu, "Robust collaborative nonnegative matrix factorization for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 10, pp. 6076–6090, Oct. 2016.
[54] C. J. Willmott and K. Matsuura, "Advantages of the mean absolute error (MAE) over the root mean square error (RMSE) in assessing average model performance," *Climate Res.*, vol. 30, no. 1, p. 79, Dec. 2005.



Zhuang Li (Student Member, IEEE) received the B.S. degree from the College of Information Science and Engineering, Xinjiang University, Urumqi, China, in 2018, and the M.S. degree from the School of electronic and Information Engineering, Harbin Institute of Technology, Harbin, China, in 2020, where he is pursuing the Ph.D. degree.

His research interests include hyperspectral anomaly detection, machine learning, and pattern recognition.



Ye Zhang (Member, IEEE) received the B.S. degree in communication engineering and the M.S. and Ph.D. degrees in communication and electronic system from the Harbin Institute of Technology, Harbin, China, in 1982, 1985, and 1996, respectively.

He is a Professor and Doctoral Supervisor of the Department of Information and Communication Engineering, Harbin Institute of Technology. His research interests include remote sensing image analysis, image/video compression and transmission, and multisource information collaboration processing.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 04:01:16 UTC from IEEE Xplore. Restrictions apply.